Chenyu Gao

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Open to relocation within the U.S.

Summary

Applied ML scientist and ECE PhD candidate at Vanderbilt University (expected 2026), with industry experience at TikTok (ByteDance) and insitro spanning multimodal LLM post-training for video understanding, reward modeling for video generation, and multimodal representation learning. Author of 40+ peer-reviewed publications, including 8 first/co-first-author papers, and inventor on a pending U.S. patent application.

Experience

Machine Learning Scientist Intern — San Jose, CA
TikTok (ByteDance) — *Jan 2026 – May 2026*

- **Latent-space reward modeling for video generation:** Developed a DiT-native reward model for spatiotemporal consistency and multi-object interaction fidelity, with a *Semantic Adapter* mapping VAE latents to semantic features distilled from a vision foundation model.
- Proposed early-step reward evaluation directly on VAE latents, exploiting early layout stabilization to reduce DiT denoising steps before scoring by 75% and eliminate VAE decoding.
- Achieved stronger agreement with human ratings than large VLM judges on target video qualities; used the reward model for GRPO-style RL post-training of video generation models.
- **LLM-driven video scoring:** Built an SFT pipeline to adapt a base LLM for multidimensional video-quality scoring, distilling rubric-guided rationales into concise outputs for higher-QPS serving.
- Built a pipeline supplying LLM-derived video-quality features to production recommendations, contributing to a 4.15% relative increase in feature penetration and platform-wide lifts in publish frequency (+0.03%) and search engagement (+0.019%).

Data Science and Machine Learning Intern — South San Francisco, CA
insitro — *Jun 2025 – Aug 2025*

- **Image–genetics self-supervised learning:** Designed a multimodal SSL framework for target discovery, combining masked image modeling with SNP-guided contrastive alignment from whole-genome sequencing data.
- **LLM-assisted genomic feature selection:** Built a human-in-the-loop LLM/ontology pipeline that reduced 1B+ SNPs from whole-genome sequencing to a curated panel of hundreds, using published genetic associations to prioritize image-observable phenotypes and make multimodal SSL computationally tractable.
- Converted learned image embeddings into disease-relevant quantitative phenotypes via regression and clustering, enabling GWAS in imaging cohorts to recover candidate genetic associations and surface target-discovery hypotheses.

Graduate Research Assistant — Nashville, TN
Vanderbilt University — *Jul 2022 – Present*

- **Large-scale ML infrastructure:** Built containerized processing, harmonization, and QC workflows for 100,000+ MRI scans from 40+ datasets across local and Slurm/HPC environments, enabling reproducible multi-cohort ML studies.
- **Generative AI privacy and safety:** Led MRI de-identification studies evaluating the privacy–utility tradeoff of defacing methods; built a cascaded diffusion-model red-team pipeline that reconstructs high-resolution 3D facial geometry from defaced head MRI, exposing re-identification risk and motivating privacy safeguards for medical data sharing.
- **Multimodal predictive modeling:** Developed brain-age models using diffusion and structural MRI from 13,398 participants across 12 datasets to support early detection of neurodegenerative disease; this work resulted in a U.S. patent application.
- **Generative modeling:** Co-developed conditional GAN models to extend incomplete MRI fields of view, rescuing otherwise unusable scans and increasing effective dataset size by 10%; also co-developed a diffusion model for joint modeling of imaging and clinical metadata.

Graduate Research Assistant — Baltimore, MD
Johns Hopkins University — *Dec 2020 – May 2022*

- **Continual learning:** Extended a lifelong learning algorithm from vision to speech and validated omnidirectional knowledge transfer on a spoken-digit benchmark, contributing to an open-source library (link) and a *TPAMI* paper.
- **Medical image analysis:** Benchmarked deep learning and computer vision methods for MRI analysis, establishing baselines for subsequent work on MRI privacy, defacing, and data utility evaluation.

Education

Vanderbilt University — Nashville, TN
Doctor of Philosophy in Electrical & Computer Engineering — *Jul 2022 – Dec 2026 (expected)*

Johns Hopkins University — Baltimore, MD
Master of Science in Biomedical Engineering — *Aug 2020 – May 2022*

Sun Yat-sen University — Guangzhou, China
Bachelor of Science in Biomedical Engineering — *Aug 2016 – Jun 2020*

Selected Publications & Patent

Full list: 40+ peer-reviewed publications, including 8 first/co-first-author works; see Google Scholar.

S1. **C Gao***, K Xu*, et al. "Pitfalls of defacing whole-head MRI: re-identification risk with diffusion models and compromised research potential." *Computers in Biology and Medicine*. 2025. **[Diffusion-model privacy red-team; generative AI safety]**

S2. **C Gao**, et al. "Brain age identification from diffusion MRI synergistically predicts neurodegenerative disease." *Imaging Neuroscience*. 2025. **[Multimodal predictive modeling; related U.S. patent application]**

S3. **Patent: C Gao**, BA Landman, ME Kim. "System and Method of Brain Age Identification for Predicting Neuro-Degenerative Disease." U.S. Patent Application No. 19/345,706. **[Lead inventor; applied predictive ML]**

S4. **C Gao**, et al. "Field-of-view extension for brain diffusion MRI via deep generative models." *Journal of Medical Imaging*. 2024. **[Conditional generative modeling]**

S5. Y Liu, **C Gao**, et al. "MetaVoxel: Joint Diffusion Modeling of Imaging and Clinical Metadata." *Medical Imaging with Deep Learning*. 2026. **[Multimodal diffusion modeling]**

S6. JT Vogelstein, J Dey, …, **C Gao**, et al. "Simple Lifelong Learning Machines." *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2025. **[Continual learning]**

S7. ME Kim, **C Gao**, et al. "White matter micro- and macrostructure brain charts for the human lifespan." *Nature*. 2026. **[Population-scale ML infrastructure; normative brain-chart modeling]**